

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}$$

$$\begin{aligned}\min f(x) &= 150x_1 + 300x_2 \\ 26x_1 &\geq 13 \\ 2x_1 + x_2 &\geq 4 \\ 3x_1 + 10x_2 &\geq 12 \\ x_1, x_2 &\geq 0\end{aligned}$$

Metodo delle due fasi

Fase I

$$\begin{aligned}\min f(x) &= a_1 + a_2 + a_3 \\ 26x_1 - s_1 + a_1 &= 13 \\ 2x_1 + x_2 - s_2 + a_2 &= 4 \\ 3x_1 + 10x_2 - s_3 + a_3 &= 12 \\ x_1, x_2, s_1, s_2, s_3, a_1, a_2, a_3 &\geq 0\end{aligned}$$

Calcolo dei costi

$$c_i = c_N - c_B^T \cdot B_N^{-1}$$

Forma Standard

$$\begin{aligned}\min f(x) &= 150x_1 + 300x_2 \\ 26x_1 - s_1 &= 13 \\ 2x_1 + x_2 - s_2 &= 4 \\ 3x_1 + 10x_2 - s_3 &= 12 \\ x_1, x_2, s_1, s_2, s_3 &\geq 0\end{aligned}$$

$$c_1 = 0 - (1; 1; 1) \cdot \begin{pmatrix} 26 \\ 2 \\ 3 \end{pmatrix} = -31$$

$$c_2 = 0 - (1; 1; 1) \cdot \begin{pmatrix} 0 \\ 1 \\ 10 \end{pmatrix} = -11$$

$$c_3 = 0 - (1; 1; 1) \cdot \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix} = 1$$

$$c_4 = 0 - (1; 1; 1) \cdot \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix} = 1$$

$$c_5 = 0 - (1; 1; 1) \cdot \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix} = 1$$

$$c_6 = 1 - (1; 1; 1) \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = 0$$

$$c_7 = 1 - (1; 1; 1) \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = 0$$

$$c_8 = 1 - (1; 1; 1) \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = 0$$

$$c_9 = 0 - (1; 1; 1) \cdot \begin{pmatrix} 13 \\ 4 \\ 12 \end{pmatrix} = -29$$

Metodo del simplesso

	x_1	x_2	s_1	s_2	s_3	a_1	a_2	a_3	b
a_1	26	0	-1	0	0	1	0	0	13
a_2	2	1	0	-1	0	0	1	0	4
a_3	3	10	0	0	-1	0	0	1	12
	-31	-11	1	1	1	0	0	0	-29

Entra x_1 ed esce a_1 :

	x_1	x_2	s_1	s_2	s_3	a_1	a_2	a_3	b
a_1	26	0	-1	0	0	1	0	0	13
a_2	2	1	0	-1	0	0	1	0	4
a_3	3	10	0	0	-1	0	0	1	12
	-31	-11	1	1	1	0	0	0	-29

$$R1 \leftarrow R1 \cdot \frac{1}{26}$$

	x_1	x_2	s_1	s_2	s_3	a_1	a_2	a_3	b
a_1	1	0	$-\frac{1}{26}$	0	0	$\frac{1}{26}$	0	0	$\frac{1}{2}$
a_2	2	1	0	-1	0	0	1	0	4
a_3	3	10	0	0	-1	0	0	1	12
	-31	-11	1	1	1	0	0	0	-29

$$R2 \leftarrow R2 - R1 \cdot 2$$

$$\begin{array}{cccccc|c} 2 & 1 & 0 & -1 & 0 & 0 & 1 & 0 & 4 \\ 2 & 0 & -\frac{1}{13} & 0 & 0 & \frac{1}{13} & 0 & 0 & 1 \\ \hline 0 & 1 & \frac{1}{13} & -1 & 0 & -\frac{1}{13} & 1 & 0 & 3 \end{array}$$

↓

	x_1	x_2	s_1	s_2	s_3	a_1	a_2	a_3	b
a_1	1	0	$-\frac{1}{26}$	0	0	$\frac{1}{26}$	0	0	$\frac{1}{2}$
a_2	0	1	$\frac{1}{13}$	-1	0	$-\frac{1}{13}$	1	0	3
a_3	3	10	0	0	-1	0	0	1	12
	-31	-11	1	1	1	0	0	0	-29

$$R3 \leftarrow R3 - R1 \cdot 3$$

$$\begin{array}{cccccc|c} 3 & 10 & 0 & 0 & -1 & 0 & 0 & 1 & 12 \\ 3 & 0 & -\frac{3}{26} & 0 & 0 & \frac{3}{26} & 0 & 0 & \frac{3}{2} \\ \hline 0 & 10 & \frac{3}{26} & 0 & -1 & -\frac{3}{26} & 0 & 1 & \frac{21}{2} \end{array}$$

↓

	x_1	x_2	s_1	s_2	s_3	a_1	a_2	a_3	b
a_1	1	0	$-\frac{1}{26}$	0	0	$\frac{1}{26}$	0	0	$\frac{1}{2}$
a_2	0	1	$\frac{1}{13}$	-1	0	$-\frac{1}{13}$	1	0	3
a_3	0	10	$\frac{3}{26}$	0	-1	$-\frac{3}{26}$	0	1	$\frac{21}{2}$
	-31	-11	1	1	1	0	0	0	-29

$$R4 \leftarrow R4 + R1 \cdot 31$$

$$\begin{array}{cccccc|c} -31 & -11 & 1 & 1 & 1 & 0 & 0 & 0 & -29 \\ 31 & 0 & -\frac{31}{26} & 0 & 0 & \frac{31}{26} & 0 & 0 & \frac{31}{2} \\ \hline 0 & -11 & -\frac{5}{26} & 1 & 1 & \frac{31}{26} & 0 & 0 & -\frac{27}{2} \end{array}$$

↓

	x_1	x_2	s_1	s_2	s_3	a_1	a_2	a_3	b
x_1	1	0	$-\frac{1}{26}$	0	0	$\frac{1}{26}$	0	0	$\frac{1}{2}$
a_2	0	1	$\frac{1}{13}$	-1	0	$-\frac{1}{13}$	1	0	3
a_3	0	10	$\frac{3}{26}$	0	-1	$-\frac{3}{26}$	0	1	$\frac{21}{2}$
	0	-11	$-\frac{5}{26}$	1	1	$\frac{31}{26}$	0	0	$-\frac{27}{2}$

Entra x_2 ed esce a_3 :

	x_1	x_2	s_1	s_2	s_3	a_1	a_2	a_3	b
x_1	1	0	$-\frac{1}{26}$	0	0	$\frac{1}{26}$	0	0	$\frac{1}{2}$
a_2	0	1	$\frac{1}{13}$	-1	0	$-\frac{1}{13}$	1	0	3
a_3	0	10	$\frac{3}{26}$	0	-1	$-\frac{3}{26}$	0	1	$\frac{21}{2}$
	0	-11	$-\frac{5}{26}$	1	1	$\frac{31}{26}$	0	0	$-\frac{27}{2}$

$$R3 \leftarrow R3 \cdot \frac{1}{10}$$

	x_1	x_2	s_1	s_2	s_3	a_1	a_2	a_3	b
x_1	1	0	$-\frac{1}{26}$	0	0	$\frac{1}{26}$	0	0	$\frac{1}{2}$
a_2	0	1	$\frac{1}{13}$	-1	0	$-\frac{1}{13}$	1	0	3
a_3	0	1	$\frac{3}{260}$	0	$-\frac{1}{10}$	$-\frac{3}{260}$	0	1	$\frac{21}{20}$
	0	-11	$-\frac{5}{26}$	1	1	$\frac{31}{26}$	0	0	$-\frac{27}{2}$

$$R2 \leftarrow R2 - R3$$

$$\begin{array}{cccccc|c} 0 & 1 & \frac{1}{13} & -1 & 0 & -\frac{1}{13} & 1 & 0 & 3 \\ 0 & 1 & \frac{3}{260} & 0 & -\frac{1}{10} & -\frac{3}{260} & 0 & 1 & \frac{21}{20} \\ \hline 0 & 0 & \frac{17}{260} & -1 & \frac{1}{10} & -\frac{17}{260} & 1 & -1 & \frac{39}{20} \end{array}$$

↓

	x_1	x_2	s_1	s_2	s_3	a_1	a_2	a_3	b
x_1	1	0	$-\frac{1}{26}$	0	0	$\frac{1}{26}$	0	0	$\frac{1}{2}$
a_2	0	0	$\frac{17}{260}$	-1	$\frac{1}{10}$	$-\frac{17}{260}$	1	-1	$\frac{39}{20}$
a_3	0	1	$\frac{3}{260}$	0	$-\frac{1}{10}$	$-\frac{3}{260}$	0	1	$\frac{21}{20}$
	0	-11	$-\frac{5}{26}$	1	1	$\frac{31}{26}$	0	0	$-\frac{27}{2}$

$$R4 \leftarrow R4 + R3 \cdot 11$$

$$\begin{array}{cccccc|c} 0 & -11 & -\frac{5}{26} & 1 & 1 & \frac{31}{26} & 0 & 0 & -\frac{27}{2} \\ 0 & 11 & \frac{33}{260} & 0 & -\frac{11}{10} & -\frac{33}{260} & 0 & 11 & \frac{231}{20} \\ \hline 0 & 0 & -\frac{17}{260} & 1 & -\frac{1}{10} & \frac{277}{260} & 0 & 11 & -\frac{39}{20} \end{array}$$

↓

	x_1	x_2	s_1	s_2	s_3	a_1	a_2	a_3	b
x_1	1	0	$-\frac{1}{26}$	0	0	$\frac{1}{26}$	0	0	$\frac{1}{2}$
a_2	0	0	$\frac{17}{260}$	-1	$\frac{1}{10}$	$-\frac{17}{260}$	1	-1	$\frac{39}{20}$
x_2	0	1	$\frac{3}{260}$	0	$-\frac{1}{10}$	$-\frac{3}{260}$	0	1	$\frac{21}{20}$
	0	0	$-\frac{17}{260}$	1	$-\frac{1}{10}$	$\frac{277}{260}$	0	11	$-\frac{39}{20}$

Entra s_3 ed esce a_2 :

	x_1	x_2	s_1	s_2	s_3	a_1	a_2	a_3	b
x_1	1	0	$-\frac{1}{26}$	0	0	$\frac{1}{26}$	0	0	$\frac{1}{2}$
a_2	0	0	$\frac{17}{260}$	-1	$\frac{1}{10}$	$-\frac{17}{260}$	1	-1	$\frac{39}{20}$
x_2	0	1	$\frac{3}{260}$	0	$-\frac{1}{10}$	$-\frac{3}{260}$	0	1	$\frac{21}{20}$
	0	0	$-\frac{17}{260}$	1	$-\frac{1}{10}$	$\frac{277}{260}$	0	11	$-\frac{39}{20}$

$$R3 \leftarrow R3 + R2$$

$$\begin{array}{cccccccc|c}
0 & 1 & \frac{3}{260} & 0 & -\frac{1}{10} & -\frac{3}{260} & 0 & 1 & \frac{21}{20} \\
0 & 0 & \frac{17}{260} & -1 & \frac{1}{10} & -\frac{17}{260} & 1 & -1 & \frac{39}{20} \\
\hline
0 & 1 & \frac{1}{13} & -1 & 0 & -\frac{1}{13} & 1 & 0 & 3
\end{array}$$

↓

	x_1	x_2	s_1	s_2	s_3	a_1	a_2	a_3	b
x_1	1	0	$-\frac{1}{26}$	0	0	$\frac{1}{26}$	0	0	$\frac{1}{2}$
a_2	0	0	$\frac{17}{260}$	-1	$\frac{1}{10}$	$-\frac{17}{260}$	1	-1	$\frac{39}{20}$
x_2	0	1	$\frac{1}{13}$	-1	0	$-\frac{1}{13}$	1	0	3
	0	0	$-\frac{17}{260}$	1	$-\frac{1}{10}$	$\frac{277}{260}$	0	11	$-\frac{39}{20}$

$$R4 \leftarrow R4 + R2$$

$$\begin{array}{cccccccc|c}
0 & 0 & -\frac{17}{260} & 1 & -\frac{1}{10} & \frac{277}{260} & 0 & 11 & -\frac{39}{20} \\
0 & 0 & \frac{17}{260} & -1 & \frac{1}{10} & -\frac{17}{260} & 1 & -1 & \frac{39}{20} \\
\hline
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 10 & 0
\end{array}$$

↓

	x_1	x_2	s_1	s_2	s_3	a_1	a_2	a_3	b
x_1	1	0	$-\frac{1}{26}$	0	0	$\frac{1}{26}$	0	0	$\frac{1}{2}$
a_2	0	0	$\frac{17}{260}$	-1	$\frac{1}{10}$	$-\frac{17}{260}$	1	-1	$\frac{39}{20}$
x_2	0	1	$\frac{1}{13}$	-1	0	$-\frac{1}{13}$	1	0	3
	0	0	0	0	0	0	1	10	0

$$R2 \leftarrow R2 \cdot 10$$

	x_1	x_2	s_1	s_2	s_3	a_1	a_2	a_3	b
x_1	1	0	$-\frac{1}{26}$	0	0	$\frac{1}{26}$	0	0	$\frac{1}{2}$
s_3	0	0	$\frac{17}{26}$	-10	1	$-\frac{17}{26}$	10	-10	$\frac{39}{2}$
x_2	0	1	$\frac{1}{13}$	-1	0	$-\frac{1}{13}$	1	0	3
	0	0	0	0	0	0	1	10	0

$z = 0$ si passa alla seconda fase.

Fase II

$$\min f(x) = 150x_1 + 300x_2$$

$$x_1 - \frac{1}{26}s_1 + \frac{1}{26}a_1 = \frac{1}{2}$$

$$\frac{17}{26}s_1 - 10s_2 + s_3 - \frac{17}{26}a_1 + 10a_2 - 10a_3 = \frac{39}{2}$$

$$x_2 + \frac{1}{13}s_1 - s_2 - \frac{1}{13}a_1 + a_2 = 3$$

$$x_1, x_2, s_1, s_2, s_3, a_1, a_2, a_3 \geq 0$$

Si eliminano le variabili artificiali introdotte nella Fase I

$$\min f(x) = 150x_1 + 300x_2$$

$$x_1 - \frac{1}{26}s_1 = \frac{1}{2}$$

$$\frac{17}{26}s_1 - 10s_2 + s_3 = \frac{39}{2}$$

$$x_2 + \frac{1}{13}s_1 - s_2 = 3$$

$$x_1, x_2, s_1, s_2, s_3 \geq 0$$

Calcolo dei costi della nuova tabella:

$$c_1 = 150 - (150; 0; 300) \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = 0$$

$$c_2 = 300 - (150; 0; 300) \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = 0$$

$$c_3 = 0 - (150; 0; 300) \cdot \begin{pmatrix} -\frac{1}{26} \\ \frac{17}{26} \\ \frac{1}{13} \end{pmatrix} = -\frac{225}{13}$$

$$c_4 = 0 - (150; 0; 300) \cdot \begin{pmatrix} 0 \\ -10 \\ -1 \end{pmatrix} = 300$$

$$c_5 = 0 - (150; 0; 300) \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = 0$$

$$c_6 = 0 - (150; 0; 300) \cdot \begin{pmatrix} \frac{1}{2} \\ \frac{39}{2} \\ 3 \end{pmatrix} = -975$$

Tabella risultante:

	x_1	x_2	s_1	s_2	s_3	b
x_1	1	0	$-\frac{1}{26}$	0	0	$\frac{1}{2}$
s_3	0	0	$\frac{17}{26}$	-10	1	$\frac{39}{2}$
x_2	0	1	$\frac{1}{13}$	-1	0	3
	0	0	$-\frac{225}{13}$	300	0	-975

Entra s_1 ed esce s_3 :

	x_1	x_2	s_1	s_2	s_3	b
x_1	1	0	$-\frac{1}{26}$	0	0	$\frac{1}{2}$
s_3	0	0	$\frac{17}{26}$	-10	1	$\frac{39}{2}$
x_2	0	1	$\frac{1}{13}$	-1	0	3
	0	0	$-\frac{225}{13}$	300	0	-975

$$R2 \leftarrow R2 \cdot \frac{26}{17}$$

	x_1	x_2	s_1	s_2	s_3	b
x_1	1	0	$-\frac{1}{26}$	0	0	$\frac{1}{2}$
s_3	0	0	1	$-\frac{260}{17}$	$\frac{26}{17}$	$\frac{507}{17}$
x_2	0	1	$\frac{1}{13}$	-1	0	3
	0	0	$-\frac{225}{13}$	300	0	-975

$$R1 \leftarrow R1 + R2 \cdot \frac{1}{26}$$

$$\begin{array}{cccc|c}
1 & 0 & -\frac{1}{26} & 0 & 0 & \frac{1}{2} \\
0 & 0 & \frac{1}{26} & -\frac{10}{17} & \frac{1}{17} & \frac{39}{34} \\
\hline
1 & 0 & 0 & -\frac{10}{17} & \frac{1}{17} & \frac{28}{17}
\end{array}$$

↓

	x_1	x_2	s_1	s_2	s_3	b
x_1	1	0	0	$-\frac{10}{17}$	$\frac{1}{17}$	$\frac{28}{17}$
s_3	0	0	1	$-\frac{260}{17}$	$\frac{26}{17}$	$\frac{507}{17}$
x_2	0	1	$\frac{1}{13}$	-1	0	3
	0	0	$-\frac{225}{13}$	300	0	-975

$$R3 \leftarrow R3 - R2 \cdot \frac{1}{13}$$

$$\begin{array}{cccc|c}
0 & 1 & \frac{1}{13} & -1 & 0 & 3 \\
0 & 0 & \frac{1}{13} & -\frac{20}{17} & \frac{2}{17} & \frac{39}{17} \\
\hline
0 & 1 & 0 & \frac{3}{17} & -\frac{2}{17} & \frac{12}{17}
\end{array}$$

↓

	x_1	x_2	s_1	s_2	s_3	b
x_1	1	0	0	$-\frac{10}{17}$	$\frac{1}{17}$	$\frac{28}{17}$
s_3	0	0	1	$-\frac{260}{17}$	$\frac{26}{17}$	$\frac{507}{17}$
x_2	0	1	0	$\frac{3}{17}$	$-\frac{2}{17}$	$\frac{12}{17}$
	0	0	$-\frac{225}{13}$	300	0	-975

$$R4 \leftarrow R4 + R2 \cdot \frac{225}{13}$$

$$\begin{array}{cccc|c}
0 & 0 & -\frac{225}{13} & 300 & 0 & -975 \\
0 & 0 & \frac{225}{13} & -\frac{4500}{17} & \frac{450}{17} & \frac{8775}{17} \\
\hline
0 & 0 & 0 & \frac{600}{17} & \frac{450}{17} & -\frac{7800}{17}
\end{array}$$

↓

	x_1	x_2	s_1	s_2	s_3	b
x_1	1	0	0	$-\frac{10}{17}$	$\frac{1}{17}$	$\frac{28}{17}$
s_1	0	0	1	$-\frac{260}{17}$	$\frac{26}{17}$	$\frac{507}{17}$
x_2	0	1	0	$\frac{3}{17}$	$-\frac{2}{17}$	$\frac{12}{17}$
	0	0	0	$\frac{600}{17}$	$\frac{450}{17}$	$-\frac{7800}{17}$

$$\begin{aligned}
x_1 &= \frac{28}{17} \\
s_1 &= \frac{507}{17} \\
x_2 &= \frac{12}{17}
\end{aligned}$$

$$\min f(x) = 150x_1 + 300x_2 \rightarrow f(x) = 150 \left(\frac{28}{17} \right) + 300 \left(\frac{12}{17} \right) = \frac{4200}{17} + \frac{3600}{17} = \boxed{\frac{7800}{17}}$$